EXPLAINABLE MACHINE LEARNING MODEL FOR STUDENTS' PERFORMANCE PREDICTION.

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A PROJECT PROPOSAL SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR A B.SC. DEGREE.

SUBMITTED TO
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1. Background to the Study:

In recent years, the use of data in education has increased significantly, as schools and universities now collect large amounts of information about students. Schools now use Learning Management Systems(LMS) to store data such as attendance records, test scores, assignments, and overall academic performance. This leads to large amounts of student learning data which can be used to model student's learning behavior (Song, 2021).

However, in many schools traditional academic monitoring is still being implemented meaning a student's performance is only measured by the result of their final examination. In large class lectures, students do not usually get the engagement required for academic success until a failure has been documented. Research shows that student success and student engagement are closely linked, showing that a strong approach to student success can enhance engagement and academic performance (Xerri et al., 2018). Challenges like these overall can lead to lower retention rates and low academic success (Johnson, 2021).

To address these limitations, schools are now implementing Educational Data Mining (EDM) to help identify the hidden pattern in student records. As Yağcı et al. (2022) suggested that EDM allows researchers to find the hidden connections in student records and predict future classroom performance.In addition to this, machine learning models have shown great success in identifying these connections and predicting student performance that can be put into practice. For example these models can now utilize attributes such as mid-term scores, previous student records, mock exam scores and teacher evaluations to accurately predict student's performance (Kumar et al., 2022).

Despite the accuracy of these models, they mainly function as "black boxes" meaning they do not always provide the explanations for the predictions made (Raji et al., 2024). For instance, a teacher cannot determine the cause of a student's weakness if they attempt to evaluate them and the system indicates that the student is at risk. The incorporation of XAI technologies with these machine learning models helps with the explanation and transparency of insights provided by the model (Raji et al., 2024). Thus, the goal of this project is to develop an explainable machine learning model for student performance prediction that is linked to an web dashboard. The web dashboard will allow educator to provide their students with clear, useful insights in addition to academic prediction.

2. Statement of the problem

Effective student performance monitoring remains a problem in many institutions despite the availability of student data in Learning Management Systems. Traditional assessment such as tests, assignments, attendance records, and examination results are retrospective in nature meaning these methods measure failure after it has already occurred rather than identifying at-risk students early enough for meaningful intervention (Johnson, 2021).

With the advancement of technology, machine learning techniques have been introduced to predict students' academic performance using available student data. Despite the effectiveness of these systems, many existing machine learning models exist as "black boxes" which is difficult to interpret since the models do not usually provide the reasons for the results provided (Raji et al., 2024). Therefore, there is a need to develop an explainable machine learning model that can not only predict students' academic performance accurately but also provide understandable explanations for the predictions generated. Such a system will assist educational institutions in making better academic decisions and providing timely support to students who may be at risk academically.

3. Aims and Objectives of the Study

Aim of the Study:

The primary aim of this study is to develop explainable machine learning model for students' performance prediction.

The Objectives of the Study is to:

i. Design a supervised Random Forest classification model to predict student academic performance using the UCI Student Performance dataset.
ii. Implement the designed model into a React web dashboard integrated with a SHAP-based Explainable AI framework for transparent and explainable predictions.
iii. Evaluate the predictive performance of the developed model using standard classification metrics like accuracy, precision, recall and F1-score, identifying the most reliable measure for early academic intervention.

4. Significance of the study:

This study is significant because it directly addresses the challenge of identifying struggling students early enough for timely academic intervention. According to Almalawi et al. (2024), predictive analysis is an essential factor in educational decision making, such as providing anticipatory measures to reduce dropout and failure risk before final grades are released. Previous studies show that motivated students excel academically, stick with their studies, and have more enjoyable college experiences (Howard et al., 2021).

The system identifies students who are at risk with understandable explanations for each student's risk level through Explainable AI (XAI). These models give educators the ability to provide accurate educational assessment, enabling the institutions to make better decisions regarding resource allocations and support strategies (Alrais, 2025).

5. Justification of the study

Traditional manual monitoring methods are often insufficient for accurately predicting a student’s performance based on the data such as historical grades, attendance and other teacher’s evaluation metrics cumulated by educational institutions.

Machine learning methodologies such as ensemble models or Logistic Regression models achieved highly precise accuracy in predicting dropout risk. (Scott, 2026). Although deep learning and Artificial intelligence provide powerful analytical capabilities, they function as a "black box" that educators find untrustworthy (Raji et al., 2024). XAI, conversely provides suitable explanations for the results of the analysis generated by the models.

Finally, the development of a web-based dashboard is justified over deploying plain analytical scripts. If a predictive model requires the execution of programming commands, it is of no practical use in a school setting. The machine learning backend will be made understandable to academic staff who are not necessarily technical by implementing a specialized user interface. The system will allow academic advisors to input student data, view risk classifications, and understand the contributing factors, all without any technical background.

6. Scope of the Study

This research solely focuses on machine models that accurately predict student performance. This predictive model will be developed using past student records, continuous assessments results, final examination scores and other student metrics.

The algorithms used will be confined to supervised machine learning methods, particularly tree based models such as Random Forest, integrated with Explainable AI (XAI) frameworks like SHAP to ensure every prediction is understandable and interpretable.

The system will be deployed as a web-based application consisting of two components.The backend will be developed using FastAPI, a Python framework that serves the trained Random Forest model and SHAP values as a REST API. The frontend will be built using React, providing a professional and accessible user interface that allows academic advisors to input student data, view risk classifications, and examine the reasoning behind each prediction through interactive SHAP visualisations.

This study does not involve the collection of live student data, real time institutional deployment, or the prediction of performance across multiple academic subjects. The system will not support multi-user authentication or database storage in this initial implementation. The scope is intentionally focused on a secondary dataset sourced from Kaggle, titled "Students Performance in 2024 JAMB".

7. Literature review

7.1. Conceptual Framework:

The integration of computer science and education has resulted in the rapid growth of Educational Data Mining (EDM) and Learning Analytics. EDM and learning analytics transformed the analysis of student performance and results (Baker & Siemens, 2014).
Predictive modelling in Educational Data Mining serves as the foundation for this study. In order to produce accurate predictions that guide early academic support, supervised machine learning algorithms process student behavioral and academic indicators as input variables.

EDM concentrates on developing methods that help identify struggling students by identifying and revealing hidden patterns in student records (Kumar et al., 2022). Alachiotis et al. (2022) suggested that in recent years, EDM has developed into an approach for revealing relationships within educational datasets and predicting the outcomes of students' results. Furthermore, recent frameworks have demonstrated that implementing these machine learning predictions directly into institutional workflows significantly enhances educator's approach to student's at risk. (Ahmed et al., 2025).

Student academic achievement, represented through final grades or performance classification levels (e.g pass/fail or risk categories), is the dependent variable in this model. The model produces a risk classification for each student, indicating the likelihood of academic success or risk. These results serve as the foundation for early academic support, which allow educational institutions to recognize and support students at risk of failure before final grades are recorded. As a result, the conceptual framework creates an organized connection between student data inputs, machine learning analysis, predicted outcome and targeted academic intervention.

7.2. Empirical review:

Recent studies have shown the effectiveness of machine learning in academic prediction.
EDM is a field aimed at enhancing learning outcomes through the mining and analysis of data acquired at various phases of a student's educational experience (Scott, 2026).

Yağcı et al. (2022) stated that Educational Data Mining (EDM) allows academic advisors to discover hidden relationships in student records to accurately predict future academic performance.

Alachiotis et al. (2022) proposed that Educational Data mining has recently evolved into a reliable approach for revealing hidden patterns within educational information and predicting individual academic performance. They emphasized that a student's behavior and readiness to learn depends on a number of factors that traditional models fail to capture effectively. In a related study, Baker & Siemens (2014) noted that EDM and learning analytics have transformed the analysis of student performance prediction and outcomes. Building on this, empirical research (like that of Kumar et al., 2022) showed that supervised machine learning methods can convert these behavioral and academic factors into actual predictions, using attributes such as midterm scores and teacher's evaluation to predict outcomes with high accuracy.

Following on this, empirical research has shifted towards predictive algorithms to process these complexities. Surenthiran et al. (2021) explored the application of artificial intelligence techniques in predicting student academic Performance. Their study implemented a multi layer perceptron neural network trained using the backpropagation algorithm. The model achieved an average classification accuracy of nearly 75%, indicating the potential effectiveness of neural network approaches in modeling complex factors that determine student performance. However, despite their accuracy, these deep learning architectures require significantly larger datasets and lack interpretability, making them less suitable for this study

Machine learning techniques including Random Forests (RF), Decision Trees, and Support Vector Machines (SVM) are models frequently used in predicting student performance (Ben George et al. 2025). Convolutional Neural Networks (CNNS) and Fully Convolutional Networks (FCNs) represent deep learning architectures that have been used in recent research to analyse image transformed student behavioral data, going beyond tabular data. Rehman et al. 2025 believed that when deep learning is used in these educational analytics, models frequently outperform traditional machine learning techniques in terms of accuracy. Additionally, modern ensemble learning methods have proven highly effective in handling the diverse and sometimes imbalanced nature of educational datasets to maintain high predictive accuracy (Ray et al., 2025).

Lundberg and Lee (2017) introduced SHAP as a framework for interpreting machine learning predictions, grounding the feature of each contribution in cooperative game theory. In education, SHAP has been used to identify which student attributes such as attendance, prior grades, and study habits influence academic risk classifications the most. Ribeiro et al. (2016) proposed LIME as a complementary XAI technique, explaining individual predictions by approximating model behaviour locally. Using XAI Ujkani et al. identified at risk students based on their performance evaluations. Recent studies confirm that implementing XAI into educational prediction models improves educator trust and willingness to act on model outputs (Raji et al., 2024).

7.3 Gap of the study:

Despite the significant advancements in predictive accuracy demonstrated by Surenthiran et al. (2021) and various deep learning approaches, a critical gap remains in the practical application of these models within educational institutions. Most current research focuses on maximizing prediction accuracy, which leads to the creation of "black-box" algorithms. Despite their technical proficiency, these models limit institutional confidence and prevent targeted interventions since they do not give educators an understandable explanation for a student's risk rating.

Moreover, previous studies rarely address practical deployment beyond the model training stage, leaving these prediction tools as standalone analytical scripts that are completely unavailable to academic staff who are not technical (Raji et al., 2024). This lack of explainability is a primary reason for the widespread adoption of predictive AI in higher education, highlighting the critical need for explainable systems (Johora et al., 2025). To address these gaps, this study integrates Explainable AI (XAI) to enhance model transparency and deploys the predictive system as an interactive web dashboard, making complex EDM outputs accessible and actionable for academic staff. This approach not only improves model interpretability but also aids in timely academic interventions, potentially enhancing student success and retention.

8. Methodology

8.1 Introduction

This chapter discusses the research methodology that was adopted in the development of the explainable predictive model and its accompanying web-based dashboard. The system is implemented using Machine Learning techniques which encompasses the following: data acquisition, data preprocessing, model training, and algorithmic evaluation. The system is designed to show how Explainable AI (XAI) techniques are integrated to ensure model transparency, with the inclusion of a React based web dashboard that provides non-technical users with an accessible and intuitive experience while utilising the system.

8.2 Data collection and Description:

The data used for this study is a secondary dataset sourced from Kaggle, titled "Students Performance in 2024 JAMB". This dataset is synthetically generated based on real-world statistical distributions of the Joint Admissions and Matriculation Board (JAMB) examination, which serves as the highly standardized university tertiary matriculation exam in Nigeria. The dataset comprises 5,000 unique student records and captures a rich mix of 17 attributes spanning demographic, socio-economic, school-level, and behavioral variables.

The independent predictive features used to model student performance include:

i. Demographic and socio economic Data: Age, Gender, Socioeconomic_Status, and Parent_Education_Level.
ii. Academic Environment & Resources: Teacher Quality, Access To Learning Materials, Extra Tutorials, and IT_Knowledge.
iii. School Profile Data: School Type (Public vs. Private), School_Location (Urban vs. Rural), and Distance To School.

iv. iv:Student Behavioral Metrics: Study_Hours_Per_Week, Attendance_Rate, Parent_Involvement, and Assignments_Completed.

The unique identifier Student_ID will be excluded from model training to prevent algorithmic overfitting. The dependent target variable for the predictive model is the cumulative JAMB_Score (which ranges from 100 to 367 within the data). For the purpose of the supervised Random Forest classification framework, this continuous target score will be engineered into a binary risk classification outcome: "High Risk" (students tracking below standard national university cutoff baselines) and "Low Risk" (students tracking to comfortably pass), allowing the system to flag at-risk candidates for early educational interventions.

8.3 Data preprocessing:

Using the open-source Python pandas and scikit-learn libraries, a structured data preprocessing pipeline will be developed to guarantee the dataset is properly formatted for training the Machine Learning and Random Forest predictive model.

Firstly, the dataset will first go through a data cleaning step to verify, handle, and fix any missing, null, or abnormal values across the student metrics before data is passed to the core model.

secondly, text-based categorical data will be systematically converted into a machine-readable numerical format. Binary attributes: such as Gender (Male/Female), School_Type (Public/Private), School_Location (Urban/Rural), and Extra_Tutorials (Yes/No) will be transformed using Label Encoding into 0 and 1 values. Multi-class nominal features, such as Socioeconomic_Status (Low, Medium, High) and Parent_Education_Level, will be mapped using One-Hot Encoding or ordinal mapping. This method prevents the tree algorithms from deducing flawed mathematical connections or artificial hierarchies between different arbitrary categories.

Feature Scaling and SHAP Preservation - Algorithms like K-Nearest Neighbors or Support Vector Machines rely heavily on distance calculations and require continuous features to be normalized on the same scale. Random Forest works differently by executing decisions via splitting data at feature thresholds rather than measuring spatial distances between points;

consequently, it does not require continuous variables to be normalized. For this reason, continuous behavioral attributes such as Study_Hours_Per_Week, Attendance_Rate, and Assignments_Completed will be intentionally kept in their original form. This is a deliberate design choice for the SHAP visualizations; when values remain in their original units, the risk explanations displayed on the web dashboard are far easier for a teacher, principal, or academic advisor to read and interpret without needing a data science background.

8.4 **Model Architecture and training:**

The dataset will be divided after the preprocessing stage to enable efficient model training and objective assessment/evaluation. Eighty percent of the dataset will be used to train the algorithm, and the other twenty percent will be kept unobserved. This is a common 80/20 train-test split technique.

The Random forest approach, implemented using the Python scikit-learn module, will be used to build the study's student risk classification system. During the training phase, a large number of individual trees are built using Random Forest, a powerful ensemble learning technique. Each tree in the forest calculates an independent prediction while analysing the input features of a new student (such as absenteeism, study time and previous continuous assessments). In order to generate a final estimate of the student's performance that would be considered accurate, the algorithm combines these individual results using either majority voting for classification or averaging for regression.

8.5 Explainability / Interpretability integration:

Ensemble techniques like Random Forest offer strong predictive accuracy, they are black box models by nature and lack the transparency needed for use in educational settings. This work incorporates the SHAP (SHapley Additive exPlanations)framework, originally proposed by Lundberg and Lee (2017), into the Machine Learning workflow to close the gap between algorithmic accuracy and human understandability.

SHAP proposes a defined contribution value to each input feature for every individual prediction, based on ideas from cooperative game theory. This facilitates two separate tiers of understandability inside the implemented system. At the global level, summary visualisations will determine the cause of risk of a student's performance.

At the local level, SHAP bar charts will break down each individual student's prediction, showing exactly which inputs pushed the model toward a particular risk classification. For example, a student to be flagged high risk based on low study hours combined with poor attendance. These visualisations will be displayed directly on the React dashboard through interactive SHAP bar charts, so that academic advisors can not only act on a prediction but also understand the reasoning behind it.

8.6 Evaluation metrics:

The predictive performance of the Random Forest model will be measured against the 20% unused dataset. The Confusion Matrix will be the basic tool for this evaluation, which categorizes the model's predictions into True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

Regarding this project, a False Negative i.e the model predicts a student will pass when they are actually at risk of failing, is the most costly type of error, as it means a struggling student will receive no extra tutelage. This is why overall accuracy alone is not enough to evaluate the model fairly. The following metrics will therefore be calculated to give a more complete picture of model performance:

Accuracy: Accuracy is the simplest and most intuitive metric. It counts up all the correct predictions (both True Positives and True Negatives) and divides by the total number of predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:
Precision is the metric to prioritize when false positives are expensive. For example, a spam filter should have high precision because flagging a legitimate email as spam (a false positive) could mean a user misses something important. This answers the question: Of all the students the model flagged as high-risk, how many were actually high-risk?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity): Recall is the metric to prioritize when false negatives are expensive. For example, a medical screening test for cancer should have high recall because missing an actual case (a false negative) could cost someone their life. This is a critical metric for this study, as it answers: Of all the students who actually failed, how many did the model successfully identify beforehand?

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: The harmonic mean of Precision and Recall. This metric is especially useful for handling imbalanced datasets (e.g., if there are significantly more passing students than failing students in the historical data).

$$\text{F1-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

8.7 Hardware and software requirements:

Hardware Requirements:

i. Processor: Minimum Quad-Core Processor (Intel Core i5 or AMD equivalent) to handle the computational load of the ensemble tree generation and SHAP value calculations.
ii. Memory (RAM): 8.00 GB minimum, to accommodate the dataset in memory during the pandas preprocessing and model training phases.
iii. Storage: 256 GB Solid State Drive (SSD) for rapid data retrieval and environment execution.

Software Requirements

i. Operating System: Windows 11 (or macOS/Linux equivalent).
ii. Programming Language: Python 3.x, selected for its extensive ecosystem of data science and Machine Learning libraries.

iii. Integrated Development Environment (IDE): Visual Studio Code (VS Code) and Jupyter Notebooks for iterative script development and testing.

Core Libraries:

- scikit-learn: For implementing the Random Forest algorithm and evaluation metrics.
- pandas and numpy: For data manipulation, preprocessing, and numerical operations.
- shap: For generating the Explainable AI (XAI) visualizations.

Backend:

- FastAPI: For building the REST API that serves model predictions and SHAP values to the frontend

Frontend:

- React: For building the interactive web-based dashboard
- Recharts: For rendering interactive SHAP visualisations within the dashboard

8.8. System Architecture

The system architecture of the proposed Explainable Machine Learning model for student performance prediction follows a client–server architecture, consisting of three major components: the frontend (user interface), the backend (application server), and the machine learning module. These components work together to ensure seamless data flow, accurate predictions, and interpretable outputs.

i. Overview of the Architecture:

The system is designed to allow academic advisors to input student data through a web interface, send the data to a backend server for processing, and receive both predictions and explanations in real time.

The architecture is divided into the following layers:

- Presentation Layer (Frontend)
- Application Layer (Backend/API)
- Machine Learning Layer (Model + Explainability)

ii. Presentation Layer (Frontend)

The presentation layer is developed using React.js and serves as the user interface of the system. It is responsible for user interaction and visualization of results.

Key features include:

- A data input form for entering student details such as study time, absences, and previous grades
- A prediction display panel showing whether a student is "High Risk" or "Low Risk"
- An explainability panel displaying SHAP visualizations to interpret model decisions
- The frontend communicates with the backend through HTTP requests (REST API), ensuring a smooth and responsive user experience.

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iii. Application Layer (Backend/API)

The backend is implemented using FastAPI, which acts as the intermediary between the frontend and the machine learning model.

Its responsibilities include:

- Receiving input data from the frontend
- Performing necessary preprocessing (encoding, formatting)
- Sending processed data to the trained model for prediction
- Returning prediction results along with SHAP explanation values
- FastAPI is chosen due to its high performance, scalability, and ease of integration with Python-based machine learning libraries.

iv. Machine Learning Layer

This layer contains the core predictive and explainability components of the system.

a) Prediction Model

- A Random Forest classifier is used to predict student performance
- The model is trained on the UCI Student Performance dataset
- It outputs a classification (High Risk / Low Risk)

b) Explainability Module

- The system integrates SHAP (Shapley Additive Explanations)
- SHAP computes feature contribution values for each prediction
- These values explain how each input feature influences the final result
- This layer ensures that predictions are not only accurate but also transparent and interpretable.

iv. Data Flow in the System

- The system follows a structured data flow process:
- The user inputs student data via the React frontend
- The data is sent to the FastAPI backend via an API request
- The backend preprocesses the input data
- The processed data is passed to the trained Random Forest model
- The model generates a prediction
- SHAP computes explanation values for that prediction
- The backend sends both prediction and explanation results to the frontend
- The frontend displays the results and visualizations to the user

vi. Architecture Benefits

The chosen system architecture provides several advantages:

- Modularity: Each component (frontend, backend, model) can be developed and updated independently
- Scalability: The backend can be extended to support more users or datasets
- Usability: The web interface allows non-technical users to interact with the system easily
- Explainability: Integration of SHAP ensures transparency in predictions
- Maintainability: Clear separation of concerns simplifies debugging and upgrades

vii. Summary

In summary, the system architecture combines modern web technologies with machine learning and explainable AI techniques to create an efficient, scalable, and user-friendly predictive system. By integrating React, FastAPI, Random Forest, and SHAP within a structured client–server framework, the system ensures both high predictive performance and interpretability, making it suitable for real-world educational environments.

8.9 User Interface and deployment

User-interface Design and Deployment:

The last stage of the process involves using a web-based Graphical User Interface (GUI) created with React.js library to deploy the trained prediction model. The system follows a client-server architecture where the Python FastAPI backend serves the trained Random Forest model and SHAP values as a REST API. The dashboard will be designed with non-technical academic advisers in mind, with an emphasis on model explainability, user experience, and accessibility.

The interface architecture will comprise of the following:

i. Data Input Module (Sidebar) Teachers will enter a student's details through a simple input form on the sidebar. This will include drop-down menus for attributes such as internet access and parental education, and sliders for numerical inputs such as attendance rate.
ii. Prediction Output Panel The main section of the dashboard will display the model's prediction immediately after the inputs are entered. The result will be shown as either "Low Risk" (likely to pass) or "High Risk" (at risk of failing), with a simple colour indicator — green for low risk and red for high risk.
iii. Explainability Panel Directly beneath the prediction, the dashboard will display a SHAP bar chart showing which features most influenced the result. For example, it will visually show that a student's high absence count and low access to learning materials were the main contributors to a high-risk classification. This gives the advisor a clear reason behind every prediction, making it easier to decide on the right intervention.

9. Expected outcome

Upon successful completion of this project, the following outcomes are expected:

i. A Functional Predictive Model: A supervised Random Forest classification model trained on the 2024 JAMB dataset, capable of identifying at-risk tertiary candidates with high accuracy, precision, and recall.
ii. Explainable Predictions via SHAP: Moving away from traditional, opaque "black-box" systems, the application will leverage SHAP to transparently justify every classification.

Educational stakeholders will see precisely which behavioral or socio-economic variables contributed to a student's risk profile.

iii. An Intuitive Web Dashboard: A polished, fully decoupled React web application that allows educational staff to effortlessly check student risk levels and view explanations without any programming or command-line execution.

iv. Proactive Academic Interventions: The ultimate objective is to empower Nigerian educators with early, preventative insights, allowing them to step in with directed tutorials and resource allocations well before the actual standardized JAMB exams sit.

10. Budget

Item	**Estimated Cost (₦)**
Data Collection and Printing	15,000
Internet Subscription	10,000
Transportation	8,000
Power Supply and Electricity	12,000
System Development Tools	10,000
Documentation and Binding	10,000
Miscellaneous Expenses	5,000
Total	70,000

11. Workplan

Activities	**Duration**
Topic Selection and Approval	Week 1
Literature Review and Data Gathering	Week 2 – 3
System Analysis and Design	Week 4
Data Collection and Preprocessing	Week 5
Model Development and Training	Week 6
System Testing and Evaluation	Week 7
Documentation and Corrections	Week 8
Final Submission	Week 9

12. References

Johnson, M. (2021). *Challenges and opportunities in the first year of college: A longitudinal study*. Journal of College Student Retention: Research, Theory & Practice.

Xerri, M. J., Radford, K., & Shacklock, K. (2018). Student engagement in academic activities: A social support perspective. *Higher Education*, 75(4), 589-605.

Raji, N. R., Kumar, R. M. S., & Biji, C. L. (2024). Explainable machine learning prediction for the academic performance of deaf scholars. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3363634.

Almalawi, A., Soh, B., Li, A., & Samra, H. (2024). Predictive models for educational purposes: A systematic review. Big Data and Cognitive Computing, 8(12), 187. https://doi.org/10.3390/bdcc8120187

Alrais, A. (2025). Predicting & analyzing university success: Machine learning approaches to predict performance from socioeconomic and educational data [Master's thesis, Rochester Institute of Technology]. RIT Digital Institutional Repository.

Howard, J. L., Bureau, J., Guay, F., Chong, J. X., & Ryan, R. M. (2021). Student motivation and associated outcomes: A meta-analysis from self-determination theory. Perspectives on Psychological Science, 16(6), 1140-1153.

Scott, M. J. (2026). A machine learning based early detection and intervention system for at-risk students [Doctoral praxis, The George Washington University]. ProQuest Dissertations and Theses Global.

Kumar, S., & Ahuja, B. (2022). Analysis and prediction of student performance by using a hybrid optimized BFO-ALO based approach. International Journal on Recent and Innovation Trends in Computing and Communication, 10(2s), 75–88.

Baker, R. S., & Siemens, G. (2014). Educational data mining and learning analytics. In *The Cambridge Handbook of the Learning Sciences* (pp. 253-274). Cambridge University Press.

Surenthiran, R., et al. (2021). Artificial Intelligence methods for the prediction of academic performance. *IEEE International Conference on Educational Technology.*

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765–4774.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135–1144.

Song, X. (2021). Effective sequential learning for student performance prediction [Doctoral dissertation, Deakin University]. Deakin University Figshare Repository.

Alachiotis, N., et al. (2022). Machine learning models for predicting student performance: A systematic review. *Journal of Educational Data Mining.*

Yağcı, M., et al. (2022). Educational data mining: Predicting student academic performance using machine learning algorithms. *Smart Learning Environments, 9*(1), 1-19.

B. Ujkani, D. Minkovska, and N. Hinov, "Course Success Prediction and Early Identification of At-Risk Students Using Explainable Artificial Intelligence," Electronics, vol. 13, no. 21, p. 4157, Oct. 2024, doi: 10.3390/electronics13214157.

Ben George, E., Senthilkumar, R., Al-Junaibi, F., & Al-Shuaibi, Z. (2025). Explainable AI methods for predicting student grades and improving academic success. Journal of Information Systems Engineering and Management, 10(23s), 117–126. https://doi.org/10.52783/jisem.v10i23s.3680

Ahmed, S., Rahman, M. M., Hossain, M., & Islam, R. (2025). Explainable machine learning framework for academic performance prediction. Computers & Education: Artificial Intelligence, 6, 100145.

Johora, F. T., Hasan, M. N., Rajbongshi, A., Ashrafuzzaman, M., & Akter, F. (2025). An explainable AI-based approach for predicting undergraduate students academic performance. Array, 26, 100384.

Ray, S., Nawaz, A., & Ahmad, M. (2025). Explainable ensemble learning for student academic performance prediction. IEEE Access, 13, 24567–24580.

13. Appendix

S/N	Name (Matric no)	Task 6 Rating	Contribution
1	Fasakin David (23/2772) - Group leader	10	i. Successfully led the group and compiled all individual submissions. ii. Developed the methodology and expected outcome. iii. Edited each write-up to ensure strict formatting and a chronological flow. iv. Wrote the background to the study section.
2	Irene Eghosa Patrick (23/2670)	8.5	ii. Scope of the study iii. Literature review iii. Wrote the system architecture
3	Adeleye Omoteniola (23/2469)	8.5	i. Statement of the problem ii. Aim and objectives of the study iii. Significance of the study iv. Justification of the study v. Budget vi. Workplan